MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Civil Engineering**	**Course Type**	**PE**
Course Name	**Pre-Stressed Concrete Design**	**Course Code**	**CIE 4066**
Semester	**VII**	**Max Marks**	**50**
Date	**27-11-2024**	**Duration**	**3hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Explain how failure due to Bonding is controlled in pretensioned members.	**4**	5	1,2,12	**2**
1B	A simply supported post-tensioned PSC beam with dimensions of 400mm x 700mm and a span of 12m has an initial prestressing force of 540 kN. A uniformly distributed working load of 15 kN/m is applied along the span. Sketch the limiting zone for eccentricity at the midsection, taking into account the limit state of serviceability criteria while utilizing a parabolic cable configuration.Take loss of prestress as 20%, M-35 grade concrete and creep coefficient as 1.5.	**6**	4	1,2,8,12	**3**
2A	Analyse a simply supported Pre-tensioned beam of span 14m for the limit state of collapse under flexure. The beam is loaded with a working load of 18kN/m. The beam is pre-stressed by 30 indented wires of 7mm diameter whose characteristic strength is 1950N/mm^2. The effective cover to the centroid of pre-stressing steel is 250mm. The grade of concrete is M40. The section details are as follows: Top flange: 475mm × 100mm, Web: 125mm × 775mm, Bottom	**6**	4	1,2,8,12	**4**

	flange:150mm × 150mm, y_t = 425.33mm, I = 1.82×10^{10} mm^4.				
2B	A PSC beam of width 250mm and depth 500mm is pre-stressed using a parabolic cable, which has zero eccentricity at supports and 150mm below GCC at mid-span. The beam is simply supported over a span of 8m and carries a superimposed load of 10 kN/m over the whole span. Determine whether the beam is safe in limit state of collapse in shear according to codal provisions and design the shear reinforcement if required. M40 grade concrete is used for the beam and is prestressed with an effective prestress of 750 kN. Assume the beam to be uncracked in flexure.	**4**	4	1,2,8,12	**3**
3A	A symmetrical 'I' section of the Post-tensioned PSC beam is designed for a single-span bridge girder spanning 20m with a flange width of 500mm, overall depth of 800mm, and thickness of the flange and web of 120mm, I = 1.58 x 10^{10}. The beam is fully prestressed and M40 grade of concrete is used. The cable is parabolic with zero eccentricity at supports and 300mm eccentricity at midspan. The beam carries a UDL of 16kN/m throughout the span. The prestressing force in tendons is 800kN, and the loss of pre-stress is 16%. Sketch the stress distribution at the midsection for the working stage and check for permissible limits as per the IS code.	**5**	3	1,2,3,12	**3**
3B	The end block of a post-tensioned beam has a rectangular section of 475mm × 1025mm. It is pre-stressed using one anchor plate of size 237.5mm × 512.5mm. C.G. of the plate is 256.25mm from the top edge and 118.75 mm from the vertical edges. Effective pre-stressing force imparted is 1500kN. Grade of concrete is M40. Compute bursting tension and design the end block reinforcement using HYSD steel.	**5**	5	1,2,12	**3**
4A	A PSC beam has a span of 8m and a rectangular cross-section of a breadth of 400mm and a depth of 600mm. It is pre-stressed with an initial pre-stressing force of 500kN. The loss of pre-	**6**	4	1,2,8,12	**3**

	stress is 20% of initial pre-stress. The cable provided has zero eccentricity at the support section and linearly increases to 100mm at the midsection. The beam is of M-35 grade concrete. If a point load of 500 kN is applied at the mid-span. Determine the deflection and check for serviceability limits according to I.S.1343-1980 specifications. Take $\Phi = 1.5$.				
4B	Explain why prestressed concrete members will have relatively higher stiffness than corresponding RCC members.	**4**	2	1,2,12	**2**
5A	Design a simply supported type-1 post-tensioned PSC beam of span 11m loaded with a working load of 16kN/m. Take the initial value of the pre-stressing force as 1400kN. The trial section details as follows: Top flange: 475mm × 100mm, Web: 125mm × 775mm, Bottom flange:150mm × 150mm, y_t = 425.33mm, I = 1.82×10^{10} $mm^{4.}$. Take M40 grade concrete and loss of pre-stress as 15%.	**8**	3	1,2,3,12	**4**
5B	Explain the prestressing methods observed during a field visit to the Netravati bridge construction site. Also describe the process of tensioning the steel tendons, including the equipment used and as well as the safety measures implemented during the process.	**2**	1	1,2,12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***2***
2	***4***
3	***13***
4	***22***
5	***9***

BL	Total Marks
2	**10**
3	**28**
4	**14**
5	